

DSA 8301: Statistical Inference for Big Data

Marking Scheme

Master of Science in Data Science & Analytics

8th April 2026

Marking Scheme

Question 2

1. Let $X_1, X_2, \dots, X_n \sim f(x | \theta)$, where $\theta \in \Theta \subset \mathbb{R}^p$.

(a) **Parametric Statistical Model**

A parametric statistical model is a family of probability distributions indexed by a finite-dimensional parameter $\theta \in \Theta$:

$$\mathcal{M} = \{f(x | \theta) : \theta \in \Theta\}.$$

(2 marks)

Components:

- Sample space $\mathcal{X}$: Set of all possible values of X . (1 mark)
- Parameter space Θ : Set of all possible parameter values. (0.5 mark)
- Model $\mathcal{M}$: Family of distributions indexed by θ . (0.5 mark)

(b) **Importance of Finite-Dimensional Parameter Spaces** (2 mark)

- Makes estimation and inference computationally tractable.
- Improves efficiency of estimators (e.g., MLE).
- Parameters are interpretable.
- Helps avoid curse of dimensionality.

(1 marks each FOR ANY TWO POINTS)

(c) Given

$$f(x|\theta) = \frac{1}{\theta} e^{-x/\theta}, \quad x > 0.$$

(i) **Identify the distribution**

Exponential distribution. (1 mark)

(ii) **Specify $\mathcal{X}$ and Θ**

- Sample space: $\mathcal{X} = (0, \infty)$ (1 mark)
- Parameter space: $\Theta = (0, \infty)$ (1 mark)

(d) **Limitations of Parametric Models** (2 mark)

- Model misspecification can lead to biased results.
- Lack of flexibility in capturing complex data patterns.
- Strong assumptions may not hold.
- Sensitive to outliers.

(1 marks each FOR ANY TWO POINTS)

2. Let $X_1, X_2, \dots, X_n$ be a random sample from a population with mean μ .

(a) **Definition of a Point Estimator** A point estimator is a statistic computed from sample data used to estimate an unknown population parameter. (1 mark)

(b) **Unbiasedness of $\bar{X}$**

The sample mean is defined as:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i.$$

(1 mark)

Take expectation:

$$E(\bar{X}) = E\left(\frac{1}{n} \sum_{i=1}^n X_i\right).$$

(1 mark)

Using linearity of expectation:

$$E(\bar{X}) = \frac{1}{n} \sum_{i=1}^n E(X_i).$$

Since $E(X_i) = \mu$ for all i :

$$E(\bar{X}) = \frac{1}{n} \cdot n\mu = \mu.$$

(1 mark)

Hence, $\bar{X}$ is an unbiased estimator of μ .

(c) **Compute Point Estimate of μ** (1 mark)

Given data:

4, 6, 5, 7, 3, 5, 6, 4

Step-by-step calculation:

$$\begin{aligned}\bar{X} &= \frac{4 + 6 + 5 + 7 + 3 + 5 + 6 + 4}{8} \\ &= \frac{40}{8} = 5\end{aligned}$$

(1 mark)

3. Let

$$\bar{X} = \frac{1}{n} \sum X_i.$$

(a) (i) **Find $\mathbb{E}[\bar{X}]$ and $\text{Var}(\bar{X})$** (3 marks)

Step 1: Expectation

$$\mathbb{E}[\bar{X}] = \mathbb{E}\left[\frac{1}{n} \sum_{i=1}^n X_i\right]$$

Using linearity of expectation:

$$\mathbb{E}[\bar{X}] = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[X_i]$$

Since $\mathbb{E}[X_i] = \mu$:

$$\mathbb{E}[\bar{X}] = \frac{1}{n} \cdot n\mu = \mu$$

(1 mark)

Step 2: Variance

$$\text{Var}(\bar{X}) = \text{Var}\left(\frac{1}{n} \sum_{i=1}^n X_i\right)$$

Assuming independence:

$$\text{Var}(\bar{X}) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i)$$

Since $\text{Var}(X_i) = \sigma^2$:

$$\text{Var}(\bar{X}) = \frac{1}{n^2} \cdot n\sigma^2 = \frac{\sigma^2}{n}$$

(2 marks)

$$\mathbb{E}[\bar{X}] = \mu, \quad \text{Var}(\bar{X}) = \frac{\sigma^2}{n}$$

(ii) **Explain the role of the CLT** (2 marks)

The Central Limit Theorem (CLT) states that, for large n , the sampling distribution of the sample mean $\bar{X}$ is approximately normally distributed:

$$\bar{X} \approx N\left(\mu, \frac{\sigma^2}{n}\right)$$

regardless of the underlying population distribution (under mild conditions)

(1 mark)

Role:

(1 mark)

- Enables approximate normal-based inference (confidence intervals, hypothesis tests).
- Justifies use of normal approximations even for non-normal data.

(b) **Effect of Large Sample Sizes on Inference** (2 marks)

- Reduces variance of estimators ($\text{Var}(\bar{X}) = \sigma^2/n$).
- Improves precision (narrower confidence intervals).
- Strengthens validity of CLT approximations.
- Increases power of hypothesis tests.

4. (a) **Definition of Consistency** (2 marks)

An estimator $\hat{\theta}_n$ is said to be *consistent* for θ if:

$$\hat{\theta}_n \xrightarrow{p} \theta \quad \text{as } n \rightarrow \infty,$$

that is,

$$\forall \varepsilon > 0, \quad \mathbb{P}(|\hat{\theta}_n - \theta| > \varepsilon) \rightarrow 0.$$

(b) **Sufficient Condition Using Bias and Variance**

(5 marks)

Statement:

If

$$\text{Bias}(\hat{\theta}_n) \rightarrow 0 \quad \text{and} \quad \text{Var}(\hat{\theta}_n) \rightarrow 0,$$

then $\hat{\theta}_n$ is consistent.

Proof:

Recall:

$$\text{MSE}(\hat{\theta}_n) = \text{Var}(\hat{\theta}_n) + [\text{Bias}(\hat{\theta}_n)]^2.$$

If both bias and variance go to zero:

$$\text{MSE}(\hat{\theta}_n) \rightarrow 0.$$

Using Chebyshev's inequality:

$$\mathbb{P}(|\hat{\theta}_n - \theta| > \varepsilon) \leq \frac{\text{MSE}(\hat{\theta}_n)}{\varepsilon^2}.$$

As $n \rightarrow \infty$, $\text{RHS} \rightarrow 0$, hence:

$$\hat{\theta}_n \xrightarrow{p} \theta.$$

(c) **Using Chebyshev to Show $\text{MSE} \rightarrow 0 \Rightarrow \text{Consistency}$**

(4 marks)

Proof:

By Chebyshev's inequality:

$$\mathbb{P}(|\hat{\theta}_n - \theta| > \varepsilon) \leq \frac{\mathbb{E}[(\hat{\theta}_n - \theta)^2]}{\varepsilon^2} = \frac{\text{MSE}(\hat{\theta}_n)}{\varepsilon^2}.$$

If

$$\text{MSE}(\hat{\theta}_n) \rightarrow 0,$$

then

$$\frac{\text{MSE}(\hat{\theta}_n)}{\varepsilon^2} \rightarrow 0.$$

Hence:

$$\mathbb{P}(|\hat{\theta}_n - \theta| > \varepsilon) \rightarrow 0,$$

which implies:

$$\hat{\theta}_n \xrightarrow{p} \theta.$$

(d) **Consistency of $\bar{X}_n$**

(3 marks)

Solution:

We know:

$$\mathbb{E}[\bar{X}_n] = \mu, \quad \text{Var}(\bar{X}_n) = \frac{\sigma^2}{n}.$$

Thus:

$$\text{Bias}(\bar{X}_n) = 0, \quad \text{Var}(\bar{X}_n) \rightarrow 0 \text{ as } n \rightarrow \infty.$$

Hence:

$$\text{MSE}(\bar{X}_n) = \frac{\sigma^2}{n} \rightarrow 0.$$

Therefore:

$$\bar{X}_n \xrightarrow{p} \mu,$$

so $\bar{X}_n$ is consistent.